

Note on Sobolev regularity for p -harmonic maps

Simone Carano, Nikos Katzourakis and Roger Moser

April 8, 2026

Abstract

2020 MSC:

Key words and phrases: Vectorial Calculus of Variations in L^∞ ; higher order problems; Absolute Minimisers, Maximum Principle.

1 Introduction

Lemma 1.1. *Let $\Omega \subset \mathbb{R}^n$ an open set and $p \in [2, \infty)$. Assume that $u : \Omega \rightarrow \mathbb{R}^N$ is a p -harmonic map, i.e. $u \in W_{\text{loc}}^{1,p}(\Omega, \mathbb{R}^N)$ satisfying*

$$\operatorname{div}(|Du|^{p-2}Du) = 0 \quad \text{in } \Omega,$$

in the sense of distributions. Then, we have

$$|Du|^p \in W_{\text{loc}}^{1,2}(\Omega).$$

Proof. We observe

$$D|Du|^p = D(|Du|^{\frac{p}{2}})^2 = 2|Du|^{\frac{p}{2}}D|Du|^{\frac{p}{2}}.$$

Therefore, since $u \in C_{\text{loc}}^{1,\alpha}(\Omega, \mathbb{R}^N)$, it is enough to show that $|Du|^{\frac{p}{2}} \in W_{\text{loc}}^{1,2}(\Omega)$.

To this purpose, we can assume $p > 2$, since for $p = 2$ the result is clearly true.

Fix $U \Subset \Omega$ open and $\varepsilon > 0$, and consider the minimum problem

$$\min_{W_u^{1,p}(U, \mathbb{R}^N)} E_\varepsilon,$$

where $W_u^{1,p}(U, \mathbb{R}^N) = u + W_0^{1,p}(U, \mathbb{R}^N)$ and

$$E_\varepsilon(v) := \int_U (|Dv|^2 + \varepsilon^2)^{\frac{p}{2}} \quad \forall v \in W^{1,p}(U, \mathbb{R}^N).$$

Since E_ε is strictly convex, there exists a unique minimiser $u_\varepsilon \in W_u^{1,p}(U, \mathbb{R}^N)$. Moreover, by regularity theory [23], we have $u_\varepsilon \in C^\infty(U, \mathbb{R}^N)$.

As $\varepsilon \rightarrow 0$, $(E_\varepsilon)_\varepsilon$ is a sequence of equi-coercive and weakly lower semicontinuous functionals on $W^{1,p}(U, \mathbb{R}^N)$. Further, for every $v \in W^{1,p}(U, \mathbb{R}^N)$, $(E_\varepsilon(v))_\varepsilon$ is monotone decreasing with limit $E(v) := \int_U |Dv|^p$. Therefore, from the theory of Γ -convergence (see [4, Section 2.4]), we have that $E_\varepsilon \xrightarrow{\Gamma} E$ w.r.t. the weak topology of $W^{1,p}(U, \mathbb{R}^N)$. As a consequence, since u is the unique

minimiser of E in $W_u^{1,p}(U, \mathbb{R}^N)$, the full sequence u_ε converges to u weakly in $W^{1,p}(U, \mathbb{R}^N)$. We have also the convergence of their corresponding norms, since

$$\begin{aligned} E(u) &= \lim_{\varepsilon} E_\varepsilon(u_\varepsilon) = \lim_{\varepsilon} \int_U (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p}{2}} \\ &\geq \limsup_{\varepsilon} \int_U |Du_\varepsilon|^p \\ &\geq \liminf_{\varepsilon} \int_U |Du_\varepsilon|^p \\ &\geq \int_U |Du|^p = E(u) \end{aligned}$$

hence, the convergence of u_ε to u is actually strong, namely

$$u_\varepsilon \rightarrow u \quad \text{in } W^{1,p}(U, \mathbb{R}^N). \quad (1.1)$$

Now, we aim at deriving some uniform energy estimates for u_ε . In the following, the symbols $\cdot$ and $\langle \cdot, \cdot \rangle$ stand for the inner products in $\mathbb{R}^n$ and $\mathbb{R}^N$ respectively, while $:$ stands for the hilbert-Schmidt inner product in $\mathbb{R}^{N \times n}$.

We start by considering the Euler-Lagrange equation corresponding to E_ε :

$$\operatorname{div} \left((|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} Du_\varepsilon \right) = 0 \quad \text{in } U.$$

Since $u_\varepsilon \in C^\infty(U, \mathbb{R}^N)$, we can differentiate the previous equation with respect to x_i , obtaining

$$\begin{aligned} 0 &= \partial_i \left[\operatorname{div} \left((|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} Du_\varepsilon \right) \right] \\ &= \operatorname{div} \left((p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon) Du_\varepsilon + (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} D\partial_i u_\varepsilon \right). \end{aligned}$$

By taking the inner product with $\partial_i u_\varepsilon$, we have

$$\begin{aligned} 0 &= \left\langle \operatorname{div} \left((p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon) Du_\varepsilon + (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} D\partial_i u_\varepsilon \right), \partial_i u_\varepsilon \right\rangle \\ &= \operatorname{div} \left((p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon) \langle Du_\varepsilon, \partial_i u_\varepsilon \rangle + (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} \langle D\partial_i u_\varepsilon, \partial_i u_\varepsilon \rangle \right) - \\ &\quad - (p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon)^2 - (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |D\partial_i u_\varepsilon|^2 \end{aligned} \quad (1.2)$$

Thus, testing the previous equation with $\eta \in C_c^\infty(U)$, we get

$$\begin{aligned} &\int_U \eta^2 \left((p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon)^2 + (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |D\partial_i u_\varepsilon|^2 \right) \\ &= -2 \int_U \eta D\eta \cdot \left((p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon) \langle Du_\varepsilon, \partial_i u_\varepsilon \rangle + (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} \langle D\partial_i u_\varepsilon, \partial_i u_\varepsilon \rangle \right). \end{aligned} \quad (1.3)$$

Using the Cauchy-Schwartz inequality, the right hand side is bounded above by

$$\begin{aligned} &\frac{1}{2} \int_U \eta^2 (p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon)^2 + 2(p-2) \int_U |D\eta|^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} |Du_\varepsilon|^2 |\partial_i u_\varepsilon|^2 + \\ &\quad + \frac{1}{2} \int_U \eta^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |D\partial_i u_\varepsilon|^2 + 2 \int_U |D\eta|^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |\partial_i u_\varepsilon|^2, \end{aligned}$$

which in turn, using $|Du_\varepsilon|^2 \leq (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{1}{2}}$, is bounded by

$$\begin{aligned} & \frac{1}{2} \int_U \eta^2 \left((p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon)^2 + (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |D\partial_i u_\varepsilon|^2 \right) + \\ & + 2(p-1) \int_U |D\eta|^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |\partial_i u_\varepsilon|^2. \end{aligned}$$

This last quantity provides an estimates from above for the left hand side of (1.3), so that

$$\begin{aligned} & \frac{1}{2} \int_U \eta^2 \left((p-2) (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon)^2 + (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |D\partial_i u_\varepsilon|^2 \right) \\ & \leq 2(p-1) \int_U |D\eta|^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |\partial_i u_\varepsilon|^2. \end{aligned}$$

In particular,

$$\int_U \eta^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{2}} (Du_\varepsilon : D\partial_i u_\varepsilon)^2 \leq 4 \frac{p-1}{p-2} \int_U |D\eta|^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |\partial_i u_\varepsilon|^2.$$

Now, we observe

$$\partial_i (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p}{4}} = p (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-4}{4}} Du_\varepsilon : D\partial_i u_\varepsilon,$$

so that

$$\int_U \eta^2 \left(\partial_i (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p}{4}} \right)^2 \leq 4p^2 \frac{p-1}{p-2} \int_U |D\eta|^2 (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p-2}{2}} |\partial_i u_\varepsilon|^2.$$

For any open subset $U' \Subset U$, we can choose $\eta \in C_c^\infty(U)$ such that $\eta = 1$ on U' . Hence, by the Hölder inequality, setting $C(p, U, U') := 4p^2 \frac{p-1}{p-2} \|D\eta\|_{L^\infty(U)}^2$, we obtain

$$\int_{U'} \left(\partial_i (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p}{4}} \right)^2 \leq C(p, U, U') E_\varepsilon(u_\varepsilon)^{\frac{p-2}{p}} \|\partial_i u_\varepsilon\|_{L^p(U)}^2. \quad (1.4)$$

Thanks to (1.1), we have $(|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p}{4}} \rightarrow |Du|^{\frac{p}{2}}$ in $L^2(U)$. Furthermore, by recalling that $E_\varepsilon(u_\varepsilon) \rightarrow E(u) = \|Du\|_{L^p(U)}^p$ as $\varepsilon \rightarrow 0$, we can pass to the limit in (1.4) and conclude

$$\left\| \partial_i |Du|^{\frac{p}{2}} \right\|_{L^2(U')}^2 \leq \liminf_{\varepsilon \rightarrow 0} \int_{U'} \left(\partial_i (|Du_\varepsilon|^2 + \varepsilon^2)^{\frac{p}{4}} \right)^2 \leq C(p, U, U') \|Du\|_{L^p(U)}^p.$$

Therefore, since U' and U were arbitrary, we have just shown that $|Du|^{\frac{p}{2}} \in W_{\text{loc}}^{1,2}(\Omega)$. □

References

- [1] G. Aronsson, *Minimization problems for the functional $\sup_x F(x, f(x), f'(x))$ I, II, III*, Arkiv fur Mat. 33 - 53 (1965); 409 - 431 (1966); 509 - 512 (1969).
- [2] A. Balci, L. Behn, L. Diening, J. Storn, *Examples of p -harmonic Maps*, SIAM Journal on Mathematical Analysis **58**(1), 260-275 (2026).
- [3] E.N. Barron, R.R. Jensen, C.Y. Wang, *The Euler equation and absolute minimizers of L^∞ functionals*, Arch. Rational Mech. Anal. **157**, 255-283 (2001).

- [4] A. Braides, *A handbook of Γ -convergence*, handbook of Differential Equations: Stationary Partial Differential Equations, North-holland, **3**, 101-213 (2006).
- [5] C. Brizzi, L. De Pascale, *A property of Absolute Minimizers in L^∞ Calculus of Variations and of solutions of the Aronsson-Euler equation*, Advances in Differential Equations **28** (3/4), 287-310 (2023).
- [6] S. Carano, N. Katzourakis, R. Moser, *Existence, uniqueness and characterisation of vector-valued absolute minimisers for a second order L^∞ -variational problem*, Preprint arXiv <https://arxiv.org/abs/2504.04181> (2025).
- [7] S. Carano, R. Moser, *An L^∞ -variational problem involving the Fractional Laplacian*, Preprint arXiv <https://arxiv.org/abs/2510.14476> (2025).
- [8] B. Dutton, N. Katzourakis, *On Second-Order L^∞ Variational Problems with Lower-Order Terms*, Discrete Contin. Dyn. Syst., **49**, 1-21 (2026).
- [9] D. Gilbarg, N. Trudinger, *Elliptic Partial Differential Equations of Second Order*. Classics in Mathematics, reprint of the 1998 edition, Springer.
- [10] N. Katzourakis, *L^∞ Variational Problems for maps and the Aronsson PDE System*, J. Differential Equations **253**(7), 2123 - 2139 (2012).
- [11] N. Katzourakis, *An L^∞ regularisation strategy to the inverse source identification problem for elliptic equations*, SIAM Journal Math, Analysis **51**(2), 1349-1370 (2019).
- [12] N. Katzourakis, *An Introduction to Viscosity Solutions for Fully Nonlinear PDE with Applications to Calculus of Variations in L^∞* , Springer Briefs in Mathematics, DOI 10.1007/978-3-319-12829-0 (2015).
- [13] N. Katzourakis, *∞ -minimal submanifolds*, Proc. Amer. Math. Soc. **142**, 2797-2811 (2014).
- [14] N. Katzourakis, *On the structure of ∞ -harmonic maps*, Comm. Partial Differential Equations **39**, 2091–2124 (2014).
- [15] N. Katzourakis, R. Moser, *Existence, uniqueness and structure of second order absolute minimisers*, ARMA **231**(3), 1615-1634 (2019).
- [16] N. Katzourakis, R. Moser, *Minimisers of supremal functionals and mass-minimising 1-currents*, Calc. Var. **64**(26) (2025).
- [17] N. Katzourakis, R. Moser, *Variational problems in L^∞ involving semilinear second order differential operators*, ESAIM: COCV, **29**(76) (2023).
- [18] N. Katzourakis, R. Moser. *Existence, uniqueness and characterisation of local minimisers in higher order Calculus of Variations in L^∞* , to appear in Analysis & PDE (2026).
- [19] N. Katzourakis, T. Pryer, *On the numerical approximation of p -Biharmonic and ∞ -Biharmonic functions*, Numerical Methods for Partial Differential Equations, **35**(1), 155-180 (2018).
- [20] N. Katzourakis, T. Pryer, *2nd order L^∞ Variational Problems and the ∞ -Polylaplacian*, Advances in Calculus of Variations **13**, 115-140 (2020).

- [21] Q. Miao, C. Wang, Y. Zhou, *Uniqueness of Absolute Minimizers for L^∞ -Functionals Involving hamiltonians $h(x, p)$* , Archive for Rational Mechanics and Analysis **223** (1), 141–198 (2017).
- [22] F. Peng, C. Wang, Y. Zhou, *Regularity of absolute minimizers for continuous convex hamiltonians*, Journal of Differential Equations, Volume 274 (2021), 1115-1164.
- [23] K. Uhlenbeck, *Regularity for a class of non-linear elliptic systems*, Acta Math. **138**, 219–240 (1977).

SIMONE CARANO

Department of Mathematics and Statistics, University of Reading
 Whiteknights Campus, Pepper Lane, Reading RG6 6AX, UK
 E-mail: s.carano@reading.ac.uk

NIKOS KATZOURAKIS

Department of Mathematics and Statistics, University of Reading
 Whiteknights Campus, Pepper Lane, Reading RG6 6AX, UK
 E-mail: n.katzourakis@reading.ac.uk

ROGER MOSER

Department of Mathematical Sciences, University of Bath
 Bath BA2 7AY, UK
 E-mail: r.moser@bath.ac.uk